

BST Physics Curriculum

Multi-volume textbook deriving Standard Model + cosmology from D_{IV}^5 substrate. Casey Koons + team (Lyra, Elie, Grace, Keeper, Cal A. Brate).

Status (Sunday 2026-05-24 EOD — KEEPER DEEP-PASS + CAL/LYRA/ELIE/GRACE CASCADE COMPLETE)

All 16 volumes (Vol 00-15) at v0.3+ substantive content with Foreword + $16 \times 12 \approx 192$ chapters. Sunday 2026-05-24 deep-pass cycle: Cal #119-#127 cold-reads (9 referee logs) → Calibration #25 + #27 both promoted STANDING via 4-instance Mode 1 trigger + Cal cross-promotion → FRAMEWORK-PLUS new tier coined ad-hoc (Cal #126) → Grace lane formally expanded per Calibration #27 STANDING corollary (INV-5139) → DCCP arc compressed from “RIGOROUS claim” through honest demote → v0.6 META-vulnerability localized to 2-way Candidate A vs D discriminable choice (Toy 3522 numerical narrowing) → SP-30-1 Bell sub-Tsirelson outreach SENT (Casey EOD) starting empirical-leg 12-18 month clock.

Vol 16 Substrate Algebra (A_{sub}) + Vol 17 Substrate Engineering QUEUED as curriculum extensions (no renumbering of Vol 0-15). Triggers: Vol 16 when Lyra #322 v0.4 reaches rigorous commutation tables + 5+ Casimir invariants (~2-4 weeks); Vol 17 when 2-3 SP-30 experimental dispatches reach paper-grade + outreach happens (~1-2 months). See KEEPER_REFINEMENT_NOTES.md for 12-chapter outlines.

Methodology stack: 25 STANDING + 1 CANDIDATE + 1 META = 27 numbered layers + FRAMEWORK-PLUS new tier (per Cal Methodology Index v0.6 + Sunday Keeper acceptance of #25). 25 commits Sunday all pushed.

Original earlier scope: 16 volumes covering substrate foundation, QFT, particles, nuclear, GR/cosmology, QM, thermo/statmech, EM, classical mechanics, condensed matter, math methods, generative geometry, chemistry, biology, information theory, methodology.

Volume directories renamed to two-digit (Vol00, Vol01, ..., Vol15) for proper sort order including Foreword (Vol00_Foreword.md).

Volume directory layout

Vol	Directory	Topic	Lead
00	Vol00_Foreword.md +	Reader contract +	Keeper + Grace +
	Vol00_Substrate_Foundation/	Substrate Foundation	Lyra
01	Vol01_QFT_from_D_IV5/	Quantum Field Theory from D_{IV}^5	Lyra
02	Vol02_Particle_Physics/	Particle Physics from D_{IV}^5	Elie
03	Vol03_Nuclear_Atomic/	Nuclear and Atomic Physics	Elie + Lyra
04	Vol04_GR_Cosmology/	General Relativity and Cosmology	Lyra + Elie
05	Vol05_Quantum_Mechanics/	Quantum Mechanics from Substrate	Lyra
06	Vol06_Thermo_Stat_Mech/	Thermodynamics and Statistical Mechanics	Lyra + Elie
07	Vol07_Electromagnetism/	Electromagnetism from D_{IV}^5	Lyra
08	Vol08_Classical_Mechanics/	Classical Mechanics as Substrate Scale 2	Lyra

Vol	Directory	Topic	Lead
09	Vol09_Condensed_Matter/	Condensed Matter (BaTiO3 137-plane + B12H32 hydride T_c ~214K)	Elie
10	Vol10_Math_Methods/	Traditional Mathematical Methods	Lyra
11	Vol11_Generative_Geometry_Topology/	Bounded symmetric domains + K3 + Heegner + Monster	Lyra
12	Vol12_Chemistry/	Periodic table + bonding + spectroscopy from substrate	Elie + Lyra
13	Vol13_Biology/	Genetic code + DNA-proton siblings + evolution-as-substrate- dynamics	Elie + Lyra
14	Vol14_Information_Theory/	RS GF(128) + Koons tick + Born=Bergman + Bell sub-Tsirelson	Lyra + Keeper
15	Vol15_Methodology/	AC(0) + AC graph + audit chain + Quaker discipline + katra	Keeper + Lyra

How to read

Start with Vol 00 Foreword for reader contract and CI-companion-era framing.

Then Vol 00 Substrate Foundation: establishes D_{IV}^5 + the five integers (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$) + $N_{max}=137$, operator zoo, conservation laws, Strong-Uniqueness Theorem.

Then Vol 01 (QFT), Vol 02 (Particles), Vol 04 (GR/Cosmology) for the physicist-prestige core.

Mathematicians: start with Vol 11 (Generative Geometry) and Vol 10 (Math Methods) instead.

Every chapter has three pedagogical registers (Casey “write for 5th graders too” standing rule): - Level 1: one-sentence essence - Level 2: graduate-physicist precision - Level 3: 5th-grader accessibility paragraph

Some chapters are author-voice condensed (~500-800 words) where the apparatus is standard; expansion to full level-1/2/3 in v3.0 publication pass.

Version label

Per Casey directive (Saturday 2026-05-23): held at v0.5 chapter-grade content state during Keeper author pass. Author-pass output is v2.0 path; current chapters at v0.2 intermediate toward v2.0 destination after Cal Phase 2 cold-read + team fact-check.

Strong-Uniqueness Theorem state

v0.10.5 ENUMERATION (Friday 2026-05-22 EOD per Cal #99 reconciliation): - 11 RIGOROUSLY CLOSED criteria - 7 candidates ($C7+C9+C15+C16+C17a+C17b+C18$) - $C18 = D_{IV}^5$ Rigidity META-theorem via $T2467+T2468$; if ratifies, null-model $\leq (1/3)^{19} \approx 9 \times 10^{-10}$

Casey-named principles (8 standing)

SWPP + Five-Absence Predictions Set + Substrate Closure + Graph Forces + Integer Web + Substrate Cognition Network + D_IV⁵ Rigidity + SCMP (Sub-Tsirelson Coherence Maintenance, $1/2^N c = 1/8$ gap as Bell falsifier).

Build (PDF regeneration)

```
pandoc --pdf-engine=xelatex -H notes/bst_pdf_header.tex <chapter>.md -o <chapter>.pdf
```

Per repo PDF pipeline convention; header file at `notes/bst_pdf_header.tex`. PDFs are stale for chapters rewritten in the Keeper full author pass and will be rebuilt at v1.0 finalization.

Cross-references

Chapter files reference AC graph theorems + K-audit chain + Strong-Uniqueness Theorem state. Per Calibration #19 STANDING RULE: external-facing chapter content uses current ratified-state count, not forecast endpoint.

— Casey Koons + Lyra + Keeper + Elie + Grace + Cal A. Brate